



Experiment 3

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1. Aim: To implement **global state management** in the EcoTrack application using **React Redux (Redux Toolkit)** for managing daily environmental activity logs, including fetching data asynchronously, adding new logs, and removing existing logs.

2. Objective:

- To understand the concept of **state management using Redux**
- To implement **Redux Toolkit** for simplified Redux configuration
- To manage application-wide state using a **centralized Redux store**
- To implement **async data fetching** using createAsyncThunk
- To add and delete data from the Redux store using reducers
- To integrate Redux with React using useDispatch and useSelector
- To simulate real-time environmental activity logging
- To understand the separation of UI logic and state logic

3. Implementation / Code:

Tools & Technologies Used:-

- React.js
- Redux Toolkit
- React Redux
- JavaScript (ES6)
- VS Code
- Web Browser (Google Chrome / Firefox)

Implementation Description:-

- 1) The EcoTrack application uses Redux Toolkit to manage daily carbon emission logs globally.
- 2) A Redux store is configured using configureStore, and a logs slice is created using createSlice.

- 3) Asynchronous data fetching is implemented using createAsyncThunk to simulate an API call that loads initial activity logs.
- 4) The Logs component uses:
- 5) useSelector to access logs data and loading status from the Redux store
- 6) useDispatch to dispatch actions such as fetching logs, adding logs, and removing logs
- 7) Users can:
- 8) Add new activities with carbon emission values
- 9) View the list of logged activities
- 10) Delete any activity from the list
- 11) This approach ensures predictable state updates, better scalability, and cleaner code organization.

Sample Code Snippet:-

 App.jsx U X

experiment-3-redux > ecotrack > src >  App.jsx > [🔍] default

```
1  import Logs from './pages/Logs';
2
3
4  function App() {
5    return (
6      <div>
7        <h1>EcoTrack (Experiment 3)</h1>
8        <Logs />
9      </div>
10   );
11 }
12
13 export default App;
14
```


